

Question				Marking details	Marks available																	
					AO1	AO2	AO3	Total	Maths	Prac												
4	(a)			<table><tr><td>Umbilical artery</td><td>Umbilical vein</td></tr><tr><td>Less oxygen</td><td>More oxygen</td></tr><tr><td>More CO₂</td><td>Less CO₂</td></tr><tr><td>More urea</td><td>Less urea</td></tr><tr><td>Less nutrients/named</td><td>More nutrients/named</td></tr><tr><td>Less antibodies</td><td>More antibodies</td></tr></table>	Umbilical artery	Umbilical vein	Less oxygen	More oxygen	More CO ₂	Less CO ₂	More urea	Less urea	Less nutrients/named	More nutrients/named	Less antibodies	More antibodies	2			2		
				Umbilical artery	Umbilical vein																	
				Less oxygen	More oxygen																	
				More CO ₂	Less CO ₂																	
				More urea	Less urea																	
				Less nutrients/named	More nutrients/named																	
				Less antibodies	More antibodies																	
	(b)			Any two (x1) from: (Barrier) {against hormones/toxic substances/microorganisms/cells/mothers rhesus group/mothers antigens } (1) Protection against differences in pressure (1) Protection against mother's {immune system/antibodies} (1)	2			2														
	(c)	(i)		Maintain {concentration/diffusion} gradients/prevents equilibrium being reached(1)		1		1														
		(ii)		(Pressure difference) forces materials through (capillaries) (1)		1		1														
		(iii)		Large (surface) area for exchange.(1)		1		1														
	(d)	(i)		GGC GTA ATT CCC		1		1														
		(ii)		different nucleotide sequence at each end/one primer for each strand of DNA (1).			1	1														
		(iii)		Enables {specific/the gene} to be {amplified/to be copied/to be replicated} (1)			1	1														
		(iv)		Does not matter how {many cycles in PCR/initial quantity/number of copies made}(1) Ratio will be same (1)			2	2														

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		(v)		Any four (x1) from 1. A and B {same/similar} ratio.(1) 2. Equal {numbers/ratios} of {the gene from chromosome 21/chromosome 21} and control chromosome/1:1 ratio (1) 3. C higher proportion of {the gene from chromosome 21/chromosome 21} or example e.g. 3:2/1.5:1 (1) 4. C = Down's syndrome (1) 5. C = Trisomy chromosome 21/three copies of chromosome 21(1)			4	4		
	(e)			Any two (x1) from: (Selective) abortion may become more common (1) Moral status of foetus/right to live(1) May cause harm to foetus (1) Could result in false {positive/negative} results(1)		2		2		
				Question 4 total	4	6	8	18	0	0